

```
1 #include <iostream>
2 #include <string>
3
4
5 using namespace std;
6
7
8
9 class Rectangle {
10 private:
11     double width;
12     double length;
13
14 public:
15     Rectangle() {
16
17         width = 0.0;
18         length = 0.0;
19     }
20
21     void setWidth(double w) {
22         if (w >= 0) {
23             width = w;
24         }
25
26         else {
27             throw display(12);
28         }
29     }
30
31
32
33     void setLength(double l) {
34         if (l >= 0) {
35             length = l;
36         }
37
38         else {
39             throw display(13);
40         }
41     }
42
43
44     double getWidth()const {
45         return width;
46     }
47
48     double getLength()const {
49         return length;
```

```
50     }
51
52
53     double getArea()const {
54         return width * length;
55     }
56
57
58     int display(int e) {
59         return e;
60     }
61
62
63
64
65
66
67
68
69 };
70
71
72 int main() {
73     double length{};
74     double width{};
75
76     double area{};
77
78     Rectangle r;
79
80     /*try {
81         cout << "Enter the value for length: ";
82         cin >> length;
83
84         if (length < 0)
85         {
86             throw r.display();
87         }
88
89
90
91         cout << "enter the value for width";
92         cin >> width;
93
94         if (width < 0) {
95             throw r.display();
96         }
97
98
```

```
99         r.setWidth(width);
100         r.setLength(length);
101
102         cout << "The Area of the rectangle is: " << r.getArea() <<      ↗
            endl << endl;
103
104
105
106     }*/
107
108     cout << "Enter the value for length: ";
109     cin >> length;
110
111
112     cout << "enter the value for width";
113     cin >> width;
114
115     try {
116         r.setLength(length);
117         r.setWidth(width);
118
119         cout << "The Area of the rectangle is: " << r.getArea() <<      ↗
            endl << endl;
120     }
121
122     catch (int e) { cout << "Error Number : " << e << endl; }
123
124
125     // can you have multiple throws to a single catch0
126
127
128
129
130
131     return 0;
132 }
133
134
```